Mooketsi Vincent Magwaza

Full-Stack Engineer · Backend, Product Systems & Platform Engineering
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PROFESSIONAL SUMMARY

Product-minded full-stack engineer who turns real operational problems into working software across interface, API, data, security, deployment, and observability. Experience spans HarvestFlow's agricultural marketplace and BioAmps clean-energy systems, alongside offline-first geospatial and inventory products. Works across TypeScript, Rust, Python, PostgreSQL, Docker, CI/CD, and Grafana-backed operations.

EDUCATION

University of Botswana 2025 - Present

Computer Science | Gaborone, Botswana

EXPERIENCE

HarvestFlow - Full-Stack Engineer May 2026 - October 2026

Agricultural marketplace | React 19, TypeScript, Tauri 2, Rust/Axum, PostgreSQL, Redis

- Developed an offline-aware Botswana marketplace connecting farmers, buyers, input suppliers, and logistics providers through produce listings, server-priced orders, transport offers, trip milestones, and append-only settlement records.
- Built React/Vite/Tauri clients and a Rust/Axum API with PostgreSQL and Redis; added role-based OTP onboarding, bounded offline caches, queued GPS check-ins, an admin operations console, Docker environments, Kubernetes manifests, and Prometheus/Grafana monitoring.

BioAmps Pty Ltd - Full-Stack Engineer November 2025 - March 2026

Clean-energy software | React, TypeScript, Vite, Tailwind CSS, Supabase, IoT monitoring

- Developed the responsive BioAmps company platform covering its mission, clean-energy projects, gallery, contact, newsletter verification, and Energy Africa Conference content.
- Contributed to the Mokgosi biogas monitoring system: user and admin dashboards for sensor readings, trends, alerts, generator status, fault events, and automation controls across edge, aggregator, and cloud-facing workflows.

SELECTED PROJECT EXPERIENCE

Kori - Offline-First Wildlife Platform 2026

Product design, full-stack engineering, geospatial data, privacy, and operations

- Built four Next.js/TypeScript apps for low-connectivity safari use: a phone field app, landing site, review console, and API reference; sightings queue in IndexedDB and sync when connectivity returns.
- Implemented MapLibre/PMTiles offline maps, QR sharing, photo-metadata removal, delayed coarse location sharing, and track-only routing; a private Savuti satellite pack passed 16/16 offline checks.
- Developed a tested FastAPI/PostGIS backend with migrations, accounts, scoped API keys, uploads, reviews, trips, lifecycle jobs, Docker Compose, OpenTelemetry, Prometheus, Grafana, Tempo, backups, and CI.

Tagwise - Browser-Based Asset, Stock & Expiry Register 2026

Product strategy, React application, spreadsheet workflows, Rust services, and data integrity

- Started from a pharmacy problem: my sister needed a practical way to track batches, bulk stock, and expiry dates without buying a desktop POS subscription plus separate extensions.
- Built a responsive React/TypeScript workspace with an Excel-like register, QR/barcode scanning, labels, batch and expiry alerts, stock-takes, roles, history, conflict handling, and validated Excel/CSV import.
- Built five Axum services behind nginx with PostgreSQL, Redis, migrations, concurrency/idempotency tests, OpenAPI docs, Docker Compose, GitHub Actions, and OpenTelemetry/Prometheus/Grafana/Tempo.

TECHNICAL SKILLS

Core languages TypeScript, Rust, Python, SQL, JavaScript, Go, Java	**Product & frontend** React, Next.js, Vite, responsive UI, accessibility, IndexedDB
Backend & APIs Axum, FastAPI, REST APIs, authentication, background jobs	**Data systems** PostgreSQL, PostGIS, Redis, SQLAlchemy, sqlx, Alembic
DevOps & observability Docker, nginx, GitHub Actions, Kubernetes, OpenTelemetry, Prometheus, Grafana, Tempo, Trivy	**Native & geospatial** Tauri 2, Windows APIs, WebView2, MapLibre, PMTiles